LECTURE 3

HTML and HTML5 Fundamentals

The raw ingredients of every website: structure, text, links, images, tables, and lists.

CSC336 Web Technologies

OVERVIEW

In This Lecture

1. Understand the role of markup languages, and the anatomy of an HTML tag
2. HTML's document structure, elements, attributes, nesting rules, and void elements
3. Headings, text formatting, and why the old <font> tag is obsolete
4. <pre>/<code> and other technical text tags, plus HTML entities
5. Hyperlinks (internal vs. external), images (including image links), audio, video
6. Tables, all three list types (including nested lists), and what HTML5 added

What Is a Markup Language?

1. HyperText — text that contains links to other text or pages
2. Markup — the text is annotated with tags describing its structure
3. Language — a shared grammar every browser agrees to follow
4. HTML is NOT a programming language — no variables, loops, or conditions; it only describes structure and content

START HERE

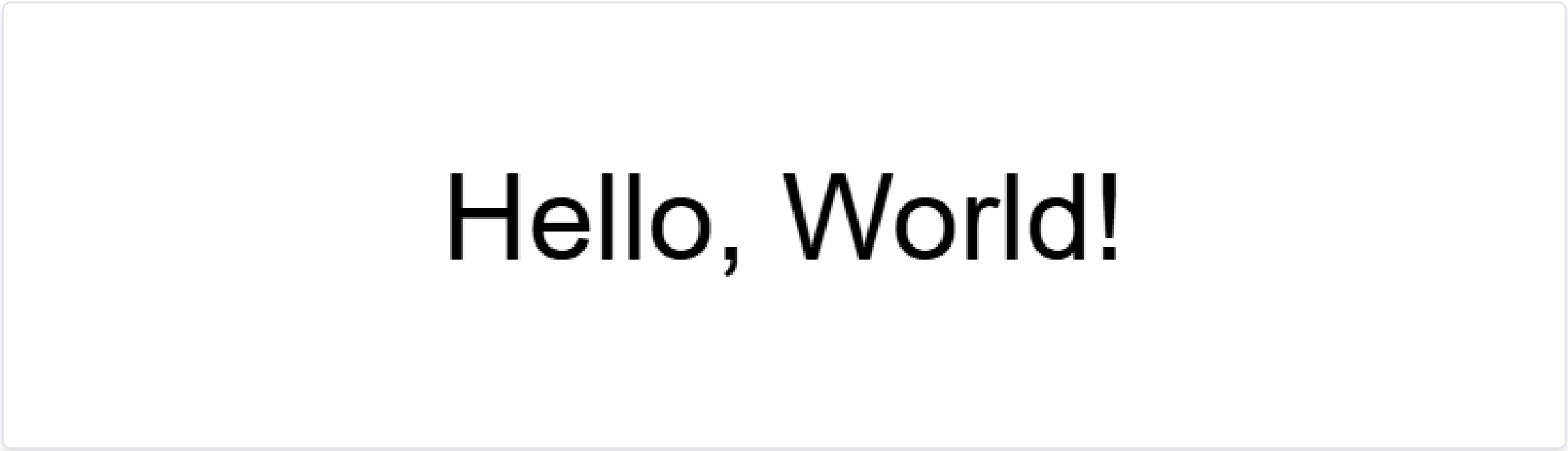
Anatomy of an HTML Tag

Part	Example	Meaning
Opening tag		Marks the start of an element
Attribute	class="intro"	Extra info, written inside the opening tag only
Content	Hello, World!	What sits between the opening and closing tag
Closing tag		Marks the end — note the forward slash
Comment	<!-- note -->	Ignored by the browser, visible only in the source
Void (self-closing) tag	 	No content, and so no closing tag at all

START HERE

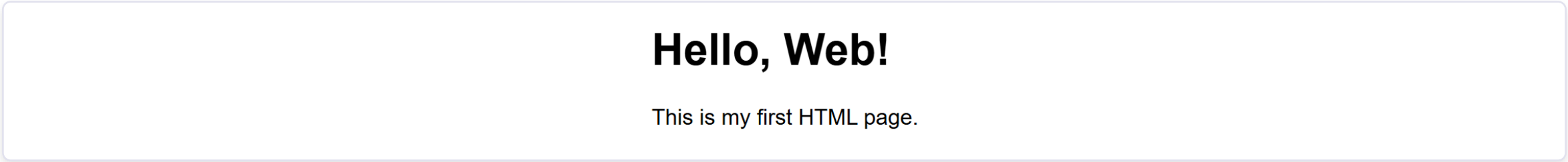
Your First Example: Hello, World!

```
<!-- This is a comment: the browser ignores it -->
<p class="intro" id="greeting">Hello, World!</p>
```



HTML Document Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>My First Page</title>
</head>
<body>
    <h1>Hello, Web!</h1>
    <p>This is my first HTML page.</p>
</body>
</html>
```



Void (Self-Closing) Elements

Element	What It Does
** **	A single line break within text
<hr>	A thematic divider line between sections
<img>	Embeds an image
<input>	A form control
<meta> / <link>	Document metadata / an external resource, used inside <head>

Void elements have no content and therefore no closing tag — is NOT void, it just happens to be empty.

Void Elements, Rendered

```
<br>
<hr>
<img src="cat.jpg" alt="A photo of a cat">
<input type="text">
```

Text before a line break.
Text after the break.

 A photo of a cat

Block-Level vs. Inline Elements

	Block-Level	Inline
Starts a new line?	Yes	No — flows with content
Full available width?	Yes, by default	No — only as wide as content
Examples	h1-h6, p, div, ul, table	a, strong, em, span, img

Block vs. Inline, Rendered

```
<h1>Page Title</h1>
<p>This is a paragraph.</p>
<p>This is another paragraph, on its own line below.</p>

<p>This word is <strong>important</strong>, and this is a
<a href="#">link</a> in the middle of a sentence.</p>
```

Page Title

This is a paragraph.

This is another paragraph, on its own line below.

This word is **important**, and this is a link in the middle of a sentence.

Global Attributes

Attribute	Purpose
id	A unique identifier for one specific element
class	Category name(s) shared by many elements, for CSS/JS
title	Extra info shown as a tooltip on hover
style	Inline CSS applied to just this one element
lang	The (human) language of this element's content
data-*	A custom attribute for attaching your own data

Boolean attributes (required, checked, disabled) need no ="value" — present means true, absent means false.

Attributes, Rendered

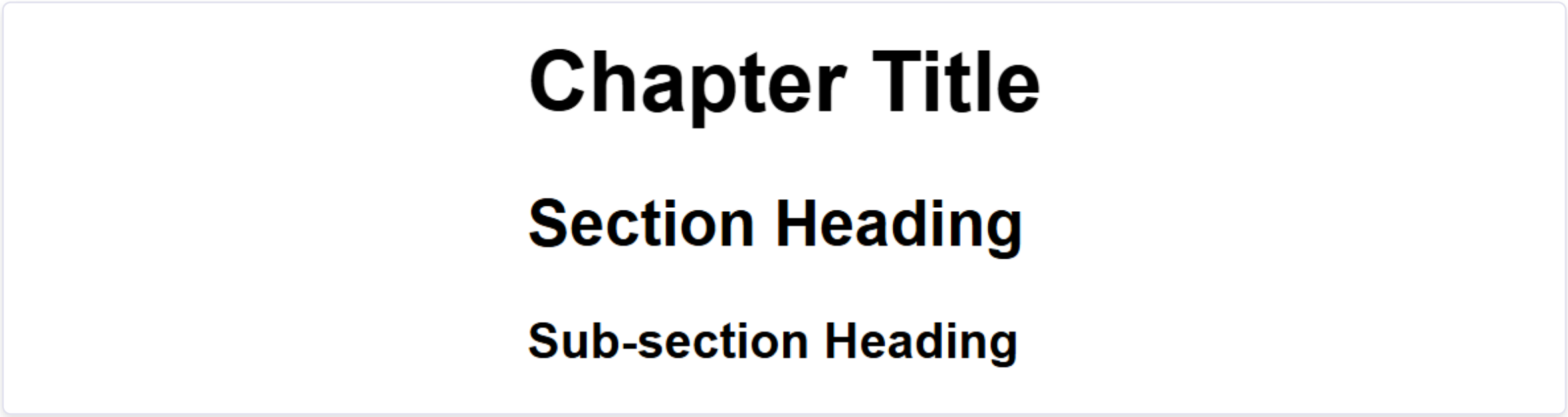
```
<a href="https://www.comsats.edu.pk" target="_blank">Visit COMSATS</a>
<p>Founded in 1998, <abbr title="World Wide Web Consortium">W3C</abbr> sets web standards.</p>
<p style="color: red;">Urgent</p>
<input type="text" required placeholder="required field">
<input type="checkbox" checked> Subscribe
<button disabled>Can't click me</button>
```



TEXT CONTENT

Headings

```
<h1>Chapter Title</h1>
<h2>Section Heading</h2>
<h3>Sub-section Heading</h3>
```



Text Formatting Elements

```
<strong>Warning</strong>: this is <em>really</em> important.
<b>Bold text</b> and <i>Italic text</i>.
<mark>highlighted</mark> text, and <small>terms apply</small>.
H<sub>2</sub>O and x<sup>2</sup>
Line one<br>Line two
<hr>
```

Warning: this is *really* important.

Bold text and *Italic text*.

highlighted text, and terms apply.

H_2O and x^2

Line one
Line two

A DEPRECATED TAG

The <font> Tag

```
<font color="red" size="5" face="Arial">
  This text is red, size 5, Arial font.
</font>
```



A DEPRECATED TAG

Why We Don't Use <font>

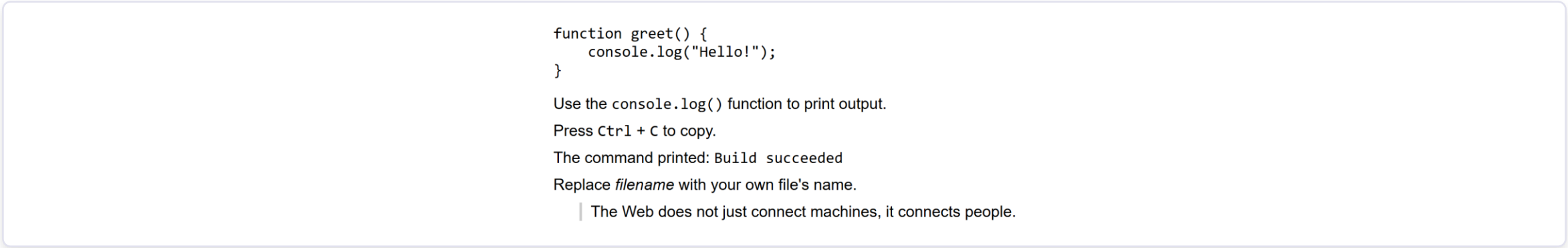
- Mixes structure with appearance — HTML should describe what something is, not how it looks
- Doesn't scale — changing 50 pages means editing 50 pages, not one CSS rule
- No responsive or interactive ability — can't change on hover, on a smaller screen, or in dark mode
- Fails validation — flagged as an error by the W3C Markup Validator

<pre>, <code>, and Related Tags

Tag	Meaning
<pre>	Preformatted text — preserves every space and line break, monospace
<code>	An inline snippet of computer code, monospace
<kbd>	Keyboard input — a key the user should press
<samp>	Sample output from a program or command
<var>	A variable name or placeholder value
<blockquote>	A longer, block-level quotation

Technical Text Tags, Rendered

```
<pre>function greet() {
    console.log("Hello!");
}</pre>
<p>Use the <code>console.log()</code> function.</p>
<p>Press <kbd>Ctrl</kbd> + <kbd>C</kbd> to copy.</p>
<p>Printed: <samp>Build succeeded</samp></p>
<p>Replace <var>filename</var> with your own.</p>
<blockquote>The Web connects people.</blockquote>
```



HTML Entities

Entity	Renders As	Why You Need It
<	<	A literal < would start a tag
>	>	A literal > would end a tag
&	&	A literal & would start another entity
** **	(a space)	A non-breaking space — never wraps a line
©	©	Not on most keyboards
—	—	An em dash, longer than a hyphen

Entities, Rendered

```
<p>5 &lt; 10 and 10 &gt; 5</p>
<p>Copyright &copy; 2025 &mdash; All rights reserved</p>
<p>Click Here Now (no line break allowed)</p>
```

5 < 10 and 10 > 5

Copyright © 2025 — All rights reserved

Click Here Now (no line break allowed between these words)

Internal and External Links

EXTERNAL (ABSOLUTE URL)

- The complete address, including https:// and the domain
- Use for a page on a DIFFERENT website
- target="_blank" + rel="noopener noreferrer" opens safely in a new tab

INTERNAL (RELATIVE URL)

- A path relative to the current page — no domain name
- Use for pages within YOUR OWN site
- Survives a domain move, since it never contained the domain

Links, Rendered

```
<a href="https://en.wikipedia.org/wiki/HTML">HTML on Wikipedia</a>
<a href="https://www.comsats.edu.pk" target="_blank" rel="noopener noreferrer">Visit COMSATS (new tab)</a>
<a href="about.html">About Us</a>
<a href="pages/contact.html">Contact (in a subfolder)</a>
<a href="../index.html">Back to home (one folder up)</a>
```

HTML on Wikipedia

Visit COMSATS (new tab)

About Us

Contact (in a subfolder)

Back to home (one folder up)

The <img> Element

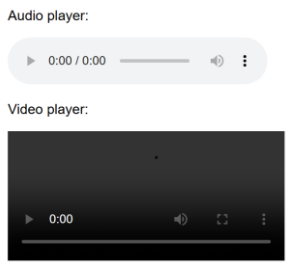
```
<img src="images/logo.png" alt="COMSATS logo" width="200" height="80">
```



Audio and Video

```
<audio controls>
    <source src="song.mp3" type="audio/mpeg">
    Your browser does not support audio.
</audio>

<video width="480" controls>
    <source src="movie.mp4" type="video/mp4">
    Your browser does not support video.
</video>
```



Tables: <table>, <tr>, <th>, <td>

```
<table>
  <thead>
    <tr><th>Name</th><th>Course</th><th>Grade</th></tr>
  </thead>
  <tbody>
    <tr><td>Ali</td><td>CSC336</td><td>A</td></tr>
    <tr><td>Sara</td><td>CSC336</td><td>A+</td></tr>
  </tbody>
</table>
```

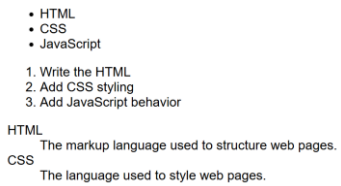
Name	Course	Grade
Ali	CSC336	A
Sara	CSC336	A+

Unordered, Ordered, and Description Lists

```
<ul>
  <li>HTML</li><li>CSS</li><li>JavaScript</li>
</ul>

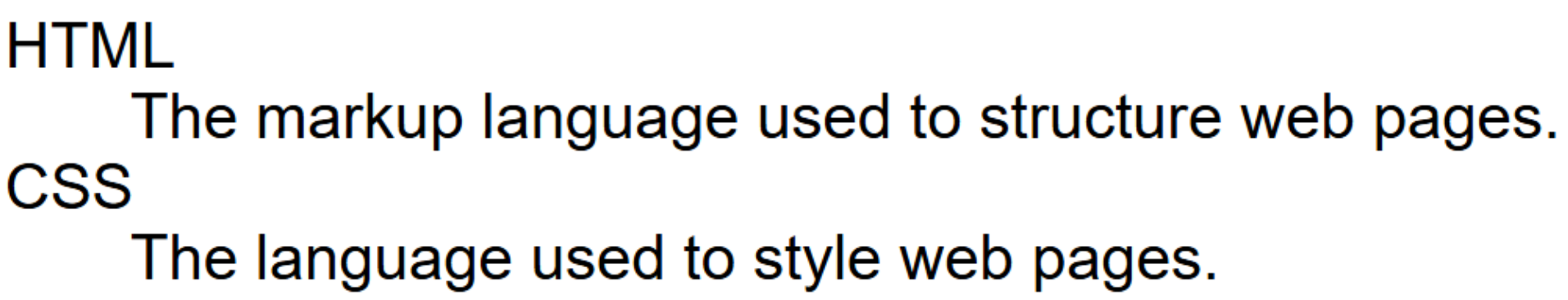
<ol>
  <li>Write the HTML</li><li>Add CSS styling</li>
</ol>

<dl>
  <dt>HTML</dt><dd>The markup language...</dd>
</dl>
```



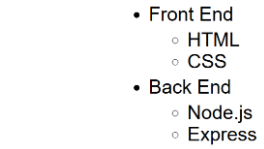
A Closer Look: Description Lists

```
<dl>
    <dt>HTML</dt>
    <dd>The markup language used to structure web pages.</dd>
    <dt>CSS</dt>
    <dd>The language used to style web pages.</dd>
</dl>
```



Nested Lists

```
<ul>
  <li>Front End
    <ul>
      <li>HTML</li>
      <li>CSS</li>
    </ul>
  </li>
  <li>Back End
    <ul>
      <li>Node.js</li>
      <li>Express</li>
    </ul>
  </li>
</ul>
```



Nested List Rules

- A nested / goes inside the <li> of the item it belongs under
- A nested <ol> restarts its own numbering from 1, independent of the level above
- Browsers change bullet style (disc → circle → square) at each nesting depth

What HTML5 Added

- Semantic layout: header, nav, main, section, article, aside, footer
- Media: audio, video, source, track — no plugins required
- Graphics: canvas and native svg support
- New attributes: placeholder, required, pattern, data-*, contenteditable, draggable

Progressive Enhancement

TIP

Not every browser supports every HTML5 feature. Check caniuse.com before relying on a newer one, and design your page so it still works reasonably well in older browsers while enhancing the experience for those that support more.

KEY TAKEAWAYS

Lecture 3 in Six Points

1. A tag has an opening tag, content, and a closing tag; attributes live inside the opening tag; comments and attributes never appear in the rendered page.
2. Void elements (br, img) have no closing tag; nesting must close in reverse order.
3. Block-level elements stack and fill width; inline elements flow within text.
4. The <font> tag is obsolete — use CSS for all styling instead.
5. <pre>/<code> handle technical text; entities (<) represent characters HTML can't take literally.
6. HTML has three list types — unordered, ordered, description (<dl>) — and any of them can be nested.